

DATA 427 Group Project: Spark ML on the KDD Dataset

Jason Pollock, Fabian Day, Bex Campbell-Redl

- 1 Implement two programs that apply Spark's Decision Tree algorithm and Logistic Regression algorithm to the provided KDD dataset. Run these programs 10 times on the school's Hadoop cluster, each using a different seed to split the dataset into a training and a test set.**

1.1 Describe the two programs using pseudo-code

1.1.1 Logistic Regression:

Algorithm 1 Spark Logistic Regression

- 1: Parse the input/output directories and number of runs
- 2: Initialise the Spark session and load the data from the provided input directory

Setup the pipeline as follows:

- 3: Index required columns
 - 4: One-hot encode required columns
 - 5: Assemble into feature columns
 - 6: Append parameterised logistic regression model as the final stage.
 - 7: **for** Number of runs **do**
 - 8: Set seed
 - 9: The data into train/test
 - 10: Fit the LR model on the train data
 - 11: Evaluate the models performance on the training and testing split, and record results
 - 12: **end for**
 - 13: Display results
-

1.1.2 Decision Tree

Algorithm 2 Spark Decision Tree

- 1: Parse the input/output directories and number of runs
 - 2: Initialise the Spark session and load the data from the provided input directory
Generate schema
Setup the pipeline as follows:
 - 3: Index required columns
 - 4: One-hot encode required columns
 - 5: Assemble into feature columns
 - 6: Append parameterised DecisionTree model as the final stage.
Run the pipeline:
 - 7: Split the test/training data into num_runs instances
 - 8: Foreach training data, train a model
Evaluate the results:
 - 9: Output the DecisionTree structure for the first tree
 - 10: Foreach (model, training data) pair, generate predictions
 - 11: Foreach (model, testing data) pair, generate predictions
 - 12: Foreach training prediction, generate an accuracy row
 - 13: Foreach test prediction, generate an accuracy row
 - 14: Produce a CSV with the aggregate data by test/train
 - 15: Produce a CSV with the per-run data
 - 16: Display results
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1.2 Describe how to install and run the two programs step by step in a Readme.txt file.

* README

group: 22

authors: Bex Campbell-Redl, Fabian Day, Jason Pollock

email: campberebe3@myvuw.ac.nz, dayfabi@myvuw.ac.nz, pollocjaso@myvuw.ac.nz

How to Install and Run Group22's Decision Tree and Logistic Regression code on the Hadoop cluster

Important note on Makefile:

Part 1: Setup & Environment

1. Ensure that you have a user directory on the HDFS cluster (if you do not have one, create one)
2. Download the submitted Tarball into a folder on your ECS desktop
3. Extract the tarball (alternatively you can clone our git repo)
4. cd into the extracted folder

Part 2: Understanding the Directory Layout

Directories:

- SparkWordCount/input - the dataset extracted and ready for use

- `SparkWordCount/bin` - where the code is placed
- `SparkWordCount/staged_output` - where local runs place their output instead of HDFS
- `DecisionTree/input` - the dataset extracted and ready for use
- `DecisionTree/bin` - where the code is placed
- `DecisionTree/staged_output` - where local runs place their output instead of HDFS
- `LogisticRegression/input` - the dataset extracted and ready for use
- `LogisticRegression/bin` - where the code is placed
- `LogisticRegression/staged_output` - where local runs place their output instead of HDFS
- `ecs_hadoop_env` - the ECS cluster Hadoop makefile variables
- `ecs_spark_env` - the ECS cluster Spark makefile variables
- `local_hadoop_env` - the Hadoop makefile variables for local execution
- `local_spark_env` - the Hadoop makefile variables for local execution

Part 3: How to run the code

The makefile supports two execution modes, `ecs` and `local`, picked via a ENV environment variable. "ecs" is the default.

The makefile will ensure proper configured access to kerberos, HDFS and Spark before attempting to launch the job.

This is configured to run via makefile. To run a variant:

Part 3a: Decision Tree

```
$ make submit_DecisionTree
```

The output will be placed in:

```
DecisionTree/output
```

```
$ cat DecisionTree/output/*.csv
```

To clean the files when complete:

```
$ make clean_DecisionTree
```

Part 3b: Logistic Regression

```
$ make submit_LogisticRegression
```

The output will be placed in:

```
LogisticRegression/output
```

```
cat LogisticRegression/output/*.csv
```

To clean the files when complete:

```
$ make clean_LogisticRegression
```

Optional Variant modes:

Runs spark wordcount

```
$ make submit_SparkWordCount
```

To Run all modes

```
$ make all
```

The output(s) will be placed in:

```
DecisionTree/output
```

```
LogisticRegression/output
```

```
SparkWordCount/output
```

```
$ cat DecisionTree/output/*.csv
```

```
$ cat LogisticRegression/output/*.csv
```

```
$ cat SparkWordCount/output/*.csv
```

To clean the files when complete:

```
$ make clean_DecisionTree
```

```
$ make clean_LogisticRegression
```

```
$ make clean_SparkWordCount
```

or

```
$ make clean
```

1.3 Report the training and test results including the max, min, average accuracy and the standard deviation obtained from the 10 runs, and the running time of each program.

1.3.1 Decision Tree:

1.3.2 Logistic Regression:

1.3.3 Comparison:

Table 1: Decision tree results for each seeded run.

Run	Seed	Train count	Test count	Train acc. (%)	Test acc. (%)
1	221	33372	14364	94.49	94.40
2	222	33414	14322	94.71	94.40
3	223	33434	14302	94.67	94.39
4	224	33373	14363	94.61	94.37
5	225	33484	14252	94.53	94.56
6	226	33351	14385	94.59	94.68
7	227	33502	14234	94.66	94.65
8	228	33167	14569	94.39	94.63
9	229	33494	14242	94.94	95.06
10	230	33482	14254	94.59	94.61

Table 2: Summary of decision tree accuracy over 10 runs.

Metric	Average	Std. dev.	Min	Max
Training accuracy (%)	94.62	0.14	94.39	94.94
Test accuracy (%)	94.57	0.20	94.37	95.06

Table 3: Logistic regression results for each seeded run.

Run	Seed	Train count	Test count	Train acc. (%)	Test acc. (%)	Time (s)
1	1	33402	14334	94.35	94.06	10.09
2	2	33429	14307	94.24	94.07	7.93
3	3	33571	14165	94.36	94.35	7.09
4	4	33434	14302	94.50	94.37	7.04
5	5	33314	14422	94.47	94.03	6.63
6	6	33570	14166	94.30	94.16	6.61
7	7	33431	14305	94.30	93.93	6.69
8	8	33535	14201	94.21	94.37	6.47
9	9	33489	14247	94.27	94.20	6.40
10	10	33357	14379	94.50	93.95	6.37

Table 4: Summary of logistic regression accuracy over 10 runs.

Metric	Average	Std. dev.	Min	Max
Training accuracy (%)	94.35	0.11	94.21	94.50
Test accuracy (%)	94.15	0.17	93.93	94.37
Elapsed time (s)	7.13	1.14	6.37	10.09

Table 5: Summary of model test accuracy and total running time over 10 runs.

Program	Average acc. (%)	Std. dev.	Min acc. (%)	Max acc. (%)	Total time (s)
Decision Tree	94.57	0.20	94.37	95.06	65.37
Spark Logistic Regression	94.15	0.17	93.93	94.37	129.29

1.4 Appendix:

1.4.1 Example of trained decision tree structure from a run

DecisionTreeClassificationModel: depth=5, numNodes=51, numClasses=2, numFeatures=41

```
if feature 4  $\leq$  30.5 then
  if feature 35  $\leq$  0.005 then
    if feature 5  $\leq$  130.5 then
      if feature 1  $\leq$  0.5 then
        Predict: 0.0
      else
        if feature 34  $\leq$  0.015 then
          Predict: 1.0
        else
          Predict: 0.0
        end if
      end if
    end if
  else
    if feature 2  $\leq$  15.5 then
      Predict: 1.0
    else
      if feature 32  $\leq$  132.5 then
        Predict: 0.0
      else
        Predict: 1.0
      end if
    end if
  end if
else
  if feature 32  $\leq$  168.5 then
    Predict: 0.0
  else
    if feature 4  $\leq$  0.5 then
      if feature 2  $\leq$  52.5 then
        Predict: 1.0
      else
        Predict: 0.0
      end if
    else
      if feature 36  $\leq$  0.015 then
        Predict: 1.0
      else
        Predict: 0.0
      end if
    end if
  end if
end if
else
  if feature 39  $\leq$  0.005 then
    if feature 1  $\leq$  1.5 then
      if feature 0  $\leq$  29.5 then
```

```

    Predict: 1.0
else
    if feature 2  $\leq$  17.5 then
        Predict: 0.0
    else
        Predict: 1.0
    end if
end if
else
    if feature 4  $\leq$  195.5 then
        if feature 30  $\leq$  0.685 then
            Predict: 1.0
        else
            Predict: 0.0
        end if
    else
        if feature 4  $\leq$  1518.5 then
            Predict: 0.0
        else
            Predict: 1.0
        end if
    end if
end if
else
    if feature 36  $\leq$  0.005 then
        if feature 4  $\leq$  14714.5 then
            if feature 5  $\leq$  608.5 then
                Predict: 0.0
            else
                Predict: 1.0
            end if
        else
            if feature 33  $\leq$  0.615 then
                Predict: 1.0
            else
                Predict: 0.0
            end if
        end if
    end if
else
    if feature 0  $\leq$  2.5 then
        if feature 7  $\leq$  0.5 then
            Predict: 1.0
        else
            Predict: 0.0
        end if
    else
        if feature 5  $\leq$  130.5 then
            Predict: 0.0
        else
            Predict: 1.0
        end if
    end if
end if

```

```
        end if
      end if
    end if
  end if
```